C Programming Fundamentals

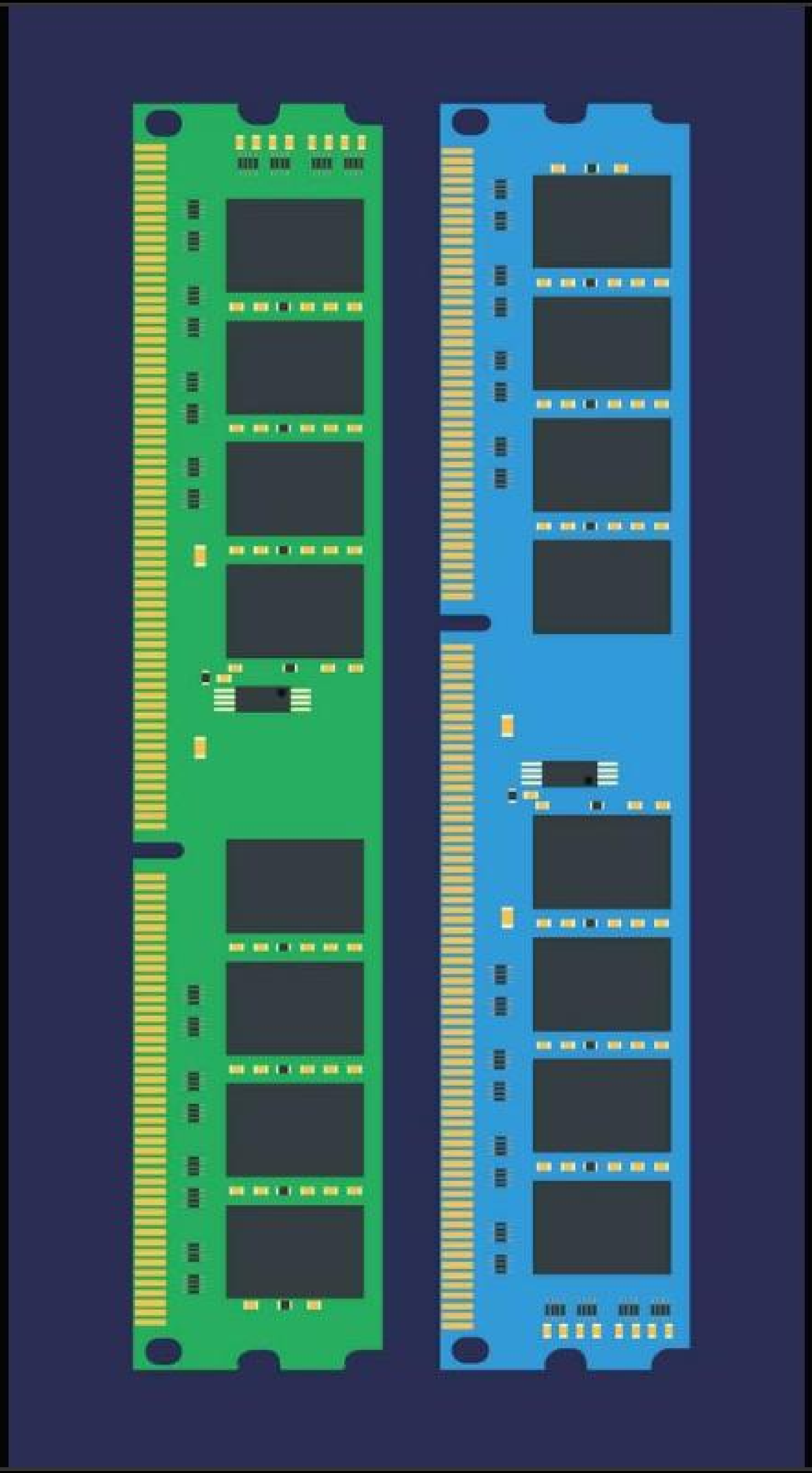
A Comprehensive Guide to Modern C Syntax and Concepts

UNDERSTANDING VARIABLES

A **Variable** is the name of a memory location which stores some data.

Think of it as a container in your computer's RAM. Each variable has a specific type and a unique name (identifier).

```
int age = 25;
```



NAMING CONVENTIONS

- **Case Sensitive:** `age` and `Age` are treated as two different variables.
- **First Character:** Must be an alphabet or an underscore (`_`). It cannot start with a digit.
- **No Spaces:** Blank spaces or commas are not allowed within a variable name.
- **Special Symbols:** No symbol other than underscore (`_`) is permitted.

PRIMARY DATA TYPES

Data Type	Size (Bytes)	Format Specifier
char	1	`%c`
int	2 or 4	`%d`
float	4	`%f`
double	8	`%lf`

TYPES OF CONSTANTS

#

Integer

Whole numbers without decimals.
Ex: `1, 0, -5`

%

Real

Numbers with decimal points.
Ex: `3.14, -2.5`

A

Character

Single characters in single quotes.
Ex: `'a', '#'`

Programming syntax graphic

RESERVED KEYWORDS

Keywords are reserved words that have a special meaning to the compiler.

There are **32 Keywords** in the standard C language that cannot be used as identifiers (variable names).

THE 32 KEYWORDS LIST

auto, break, case, char	const, continue, default, do	double, else, enum, extern	float, for, goto, if
int, long, register, return	short, signed, sizeof, static	struct, switch, typedef, union	unsigned, void, volatile, while

STANDARD PROGRAM STRUCTURE

```c
#include

int main() {
    printf("Hello World");
    return 0;
}
```

Key Components:

`#include` : Preprocessor directive.

`main()` : Entry point of the program.

`printf` : Standard output function.

CODE DOCUMENTATION

Single Line

Used for short descriptions.

```
// This is a comment
```

Multi-Line

Used for longer explanations.

```
/* This is a
multi-line comment */
```

OUTPUT & FORMAT SPECIFIERS

%d

Integers

```
printf("%d", age);
```

%f

Real Numbers

```
printf("%f", pi);
```

%c

Characters

```
printf("%c", ch);
```

READING USER INPUT

The `scanf()` function is used to take input from the user.

```
scanf("%d", &age);
```

The **& (ampersand)** symbol is crucial as it refers to the memory address of the variable.

Coding input graphic

THE COMPILATION PROCESS

Source Code
Hello.c

→

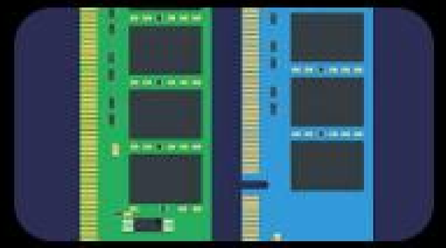
C Compiler
Translation

→

Executable
a.exe / a.out

"A computer program that translates C code into machine-executable binary code."

IMAGE SOURCES



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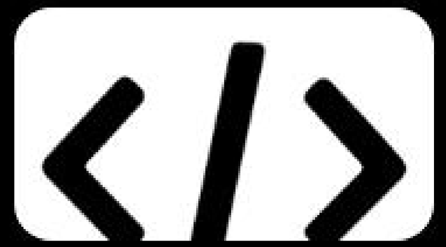

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